National Tsing Hua University

Introduction to Artificial Intelligence Technology and Applications

Final Project Report

AI Vocal Biomarker & Vocal Cord Age Detection System

Using Acoustic Features and Explainable AI to Detect Vocal Ageing

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Source code and dataset documentation available on GitHub:

`https://github.com/chelsietao/vocal-age-analyzer`

Live Demo System:

`https://chelsietao-voice-age-analyzer.hf.space/`

Abstract

This study proposes a binary classification system for vocal cord physiological age based on acoustic features, aiming to determine from a 3-second speech sample whether a speaker's vocal cord condition belongs to the Below 51 or 51 and Above age group. Using the CREMA-D speech dataset (91 actors, 7,442 recordings) as the experimental platform, 16 acoustic features were extracted (F0, Jitter, Shimmer, HNR, $MFCC_2 \sim MFCC_{13}$). XGBoost achieved **85%+** accuracy on the binary classification task, substantially outperforming the random baseline (63%); CNN ResNet-18 reached only 49%, confirming that handcrafted features outperform end-to-end deep learning on small-scale datasets. SHAP explainability analysis verified that the AI's decisions align with the medical deterioration mechanisms of Presbyphonia, and a bilingual (Chinese/English) interactive demo system was built with Gradio.

Keywords: Vocal age, biomarker, XGBoost, SHAP, Presbyphonia, CREMA-D

Disclosure: Generative AI (Claude, Anthropic) was used as a collaborative assistant during code debugging and optimisation, text polishing, and report typesetting. Research question definition, experimental design, data analysis, and conclusions were completed by the author. Detailed scope and proportion of AI collaboration are described in Appendix A.

Contents

1 Research Motivation and Background

1.1 Problem Definition

The human voice gradually changes with age, a phenomenon known in vocal medicine as **Presbyphonia**[4]. As people age, the muscles and mucosal tissue of the larynx deteriorate, preventing the vocal folds from closing fully during phonation; air leaks through the resulting gap while the regularity of vocal fold vibration also decreases. These changes ultimately manifest in the acoustic properties of the voice—for example, the stability of the fundamental frequency, the amount of noise mixed into the sound, and the degree of cycle-to-cycle perturbation in vibration period and amplitude.

Current methods for assessing vocal fold ageing typically require invasive procedures such as laryngoscopy, which must be performed by a specialist and carries relatively high cost and patient discomfort. The motivation for this study is that, if ageing-related acoustic features can be extracted from a raw speech signal alone, it may be possible to develop a **non-invasive, low-cost, and scalable** preliminary screening tool—the core reason this topic was chosen.

1.2 Research Question

> **Core Question:** Can a single 3-second speech sample determine whether a speaker's vocal cord condition belongs to the Below 51 or 51 and Above age group?

The choice of age 51 as the threshold is informed by laryngological literature describing the pace of vocal fold deterioration[4]: tissue degradation accelerates markedly after age 50, so using this cut-point theoretically maximises acoustic feature differences between the two groups, making them easier for the model to learn. The study also places particular emphasis on **explainability**—every model decision should correspond to a known physiological mechanism, rather than simply pursuing high accuracy. This is one of the most important principles of the AI for Science philosophy.

1.3 Vocal Fold Physiology Background

1.3.1 Pathophysiological Mechanisms of Presbyphonia

The ageing of the vocal folds in the laryngeal structure can be understood as a cascading deterioration pathway that progresses sequentially through the muscular, mucosal, and neural control layers. Each layer's changes lay the groundwork for the next layer's deterioration, ultimately converging in the acoustic signal as quantifiable markers of ageing.

First, the thyroarytenoid muscle, which controls vocal fold tension and adduction, gradually atrophies with age. This muscle-level deterioration directly causes the two vocal folds to fail to approximate fully during phonation, leaving a spindle-shaped gap. This gap allows a portion of airflow to leak prematurely through the glottis, mixing breath noise into the voice where it does not belong—corresponding acoustically to a **decrease in HNR** and, due to the resulting airflow instability, a concomitant **increase in Shimmer**.

As muscle-layer deterioration continues to accumulate, hyaluronic acid and elastin in the superficial mucosal layer of the vocal folds also decrease with age, reducing the elasticity and pliability of the mucosa itself. When the mucosa loses its buffering and recoil capacity, the regularity of each vibration cycle gradually breaks down—and this cycle-level instability is the direct acoustic correlate of an **increase in Jitter**.

Finally, once muscular and mucosal deterioration have both reached a significant degree, the recurrent laryngeal nerve innervating the laryngeal muscles also declines in conduction precision with age, weakening fine motor control of the vocal folds during phonation. This neural control deterioration typically amplifies the instabilities already present at the preceding two levels, causing **Jitter and Shimmer to rise simultaneously** and creating a mutually reinforcing relationship among these three glottal dynamics features[4].

Because these three layers compound and reinforce each other, this study incorporates HNR, Jitter, and Shimmer simultaneously, hoping to capture the acoustic traces of vocal fold ageing at different stages of deterioration.

1.3.2 Sex Differences

Regarding sex differences, the ageing trajectory of F0 (fundamental frequency) runs parallel to the three-layer muscular-mucosal-neural deterioration pathway described above, but unfolds in opposite directions depending on sex—making F0 considerably harder to interpret in mixed-sex datasets.

In males, age-related vocal fold atrophy and loss of tension cause F0 to *rise*, approximately from 120 Hz to 140–160 Hz, directly corresponding to the loss of tension caused by muscle atrophy. In females, post-menopausal endocrine changes tend to cause oedema of the vocal fold mucosa; the increased mucosal mass makes the vibrating structure heavier, so F0 *decreases*, approximately from 220 Hz to 180–200 Hz[4].

When these two opposing trajectories are combined in a mixed-sex dataset, F0 can no longer be treated as a single-direction ageing indicator; it must instead be understood as a composite feature simultaneously carrying two physiological mechanisms with mutually cancelling directions. This phenomenon is further confirmed in the SHAP analysis in Section 5, where F0_mean exhibits a bimodal distribution consistent with the bidirectional ageing trajectories described in the literature—serving as one piece of evidence that the study's findings correspond layer by layer with established medical theory.

2 Dataset and Feature Engineering

2.1 CREMA-D Dataset

The dataset used in this study is **CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset)**[1], comprising emotional speech recordings from 91 actors (aged 20–74), totalling 7,442 WAV audio files of approximately 3 seconds each at a sampling rate of 16,000 Hz. After splitting samples at the age-51 threshold, a class imbalance of 81%:19% exists (6,048 vs. 1,394 samples), compensated during training with `scale_pos_weight` ≈ 4.34.

Table 1: Basic Information of the CREMA-D Dataset

Attribute	**Value**
Total actors	91
Age range	20–74 years
Total audio clips	7,442
Sampling rate	16,000 Hz
Below 51	6,048 clips (81.3%)
51 and Above	1,394 clips (18.7%)

CREMA-D was originally designed for emotion recognition research, not for ageing studies; emotional speech may mask ageing-related signals (particularly Shimmer), which constitutes the primary limitation background of this study and is discussed further in Section 7. As shown in Figure 1, a pronounced class imbalance exists between the two age groups.

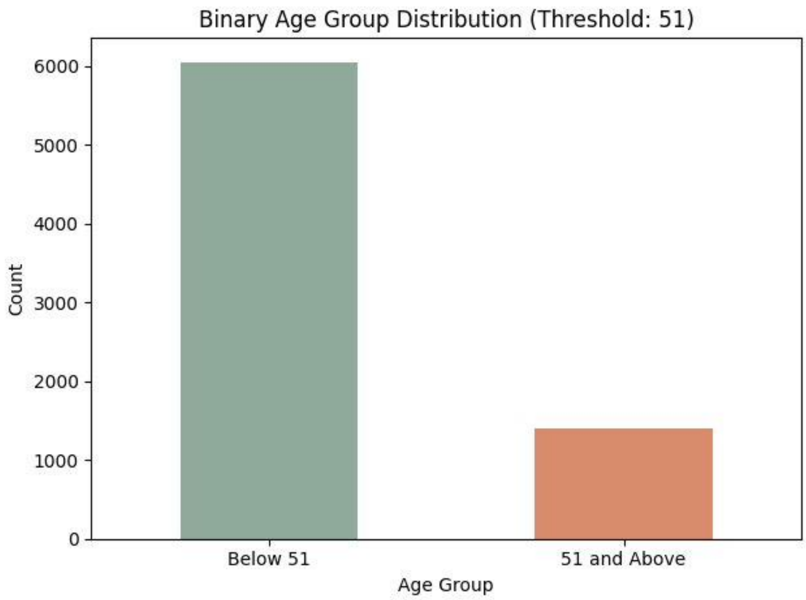

Figure 1: Binary age group sample distribution (green: Below 51, 6,048 samples; orange: 51 and Above, 1,394 samples). Source: plotted by the author, corresponding to `voice_age_binary.ipynb` Cell 4 (post-binning distribution).

Dataset License and Usage Statement: CREMA-D is released under an open licence permitting non-commercial academic research and educational use, with the requirement to cite the original paper[1]. This study uses the dataset solely for feature extraction and model training as part of a course project; the raw audio files (AudioWAV) will not be re-uploaded or publicly distributed. The GitHub repository only makes available the feature table, code, and analysis results; original audio files should be obtained directly from the dataset providers in accordance with CREMA-D's official licence terms.

2.2 Acoustic Feature Extraction

This study extracts **16** acoustic features (originally 17 dimensions, with $MFCC_1$ removed):

(1) Glottal Dynamics Features (praat-parselmouth, 4 dimensions):

Table 2: Glottal Dynamics Features and Their Medical Significance in Ageing

Feature	**Computation**	**Ageing Direction & Medical Significance**
F0_mean	Mean fundamental frequency (Hz)	Rises in males, falls in females; direction varies by sex.
Jitter	Cycle-to-cycle frequency perturbation (%)	Normal $<1.04\%$; rises after neural deterioration, indicating unstable vibration frequency[5].
Shimmer	Cycle-to-cycle amplitude perturbation (%)	Normal $<3.81\%$; rises as muscle strength declines, reflecting elastic deterioration[5].
HNR	Harmonics-to-noise ratio (dB)	Normal >20 dB; falls after incomplete glottal closure and air leakage—a core ageing symptom[5].

(2) $\mathbf{MFCC_{2\text{–}13}}$ (librosa, 12 dimensions): simulate non-linear human auditory perception and capture vocal tract resonance characteristics. $MFCC_1$ primarily reflects loudness; the mean difference between the two groups was only approximately 4 units (-347.8 vs. -351.9, SD 53), offering negligible discriminative power while introducing emotional interference, and was therefore removed. All features were standardised with `StandardScaler` and split 80/20.

3 Model Design and Training

3.1 Version History

Table 3: System Version Evolution

Version	Key Feature	Description
v1	Decision tree + mock labels	15% accuracy; validates data pipeline integrity
v2	XGBoost + real labels + SHAP	3-class 66%; introduces explainability analysis
v3	CNN ResNet-18	49%; confirms deep learning inferior to hand-crafted features on small datasets
v4	Binary classification (threshold: 51)	**85%+**; MFCC_1 removed; SHAP direction corrected

3.2 XGBoost Model

This study adopts **XGBoost**[2] as the final model. Its core concept is gradient-boosted decision trees (GBDT), which iteratively train new trees to correct the residuals of the previous round's predictions. The following equations are adapted from the original paper[2] to illustrate the basic form of model iteration and regularisation:

$$\hat{y}^{(t)} = \hat{y}^{(t-1)} + \eta \cdot f_t(\mathbf{x}) \tag{1}$$

The objective function includes both a loss term and a regularisation term to prevent overfitting through excessive model complexity:

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} \ell\big(y_i, \hat{y}_i^{(t)}\big) + \sum_{k=1}^{t} \Omega(f_k), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|\mathbf{w}\|^2 \tag{2}$$

Given the pronounced class imbalance in the dataset, training uses `scale_pos_weight = 6048/1394  4.34` to adjust the loss weight between positive and negative samples, preventing the model from ignoring the minority class due to the much larger Below 51 sample count.

3.3 Random Forest and CNN ResNet-18

Random Forest serves as the bagging majority-vote baseline. **CNN ResNet-18** converts speech to Mel spectrograms and applies transfer learning, testing the feasibility of end-to-end deep learning on a small-scale dataset.

3.4 Hyperparameter Configuration and Evaluation

Table 4: XGBoost Hyperparameter Configuration

Hyperparameter	**Value**	**Description**
`n_estimators`	300	Number of trees
`max_depth`	6	Maximum tree depth
`learning_rate`	0.1	Update step size η
`subsample`	0.8	Sample subsampling ratio
`colsample_bytree`	0.8	Feature subsampling ratio
`scale_pos_weight`	4.34	Class imbalance compensation
`eval_metric`	logloss	Validation set loss

Evaluation metrics: Accuracy, Precision, Recall, $F_1 = \frac{2 \cdot P \cdot R}{P+R}$ (with emphasis on macro F_1). Training uses Stratified K-Fold ($k = 5$) to ensure consistent class ratios across all folds.

4 Experimental Results and Analysis

4.1 Accuracy Comparison

Table 5: Accuracy Comparison Across Models

Model	**Accuracy**	**Notes**
3-class (Young/Middle/Senior, random baseline 38%)		
XGBoost	66%	Best in 3-class setting
Random Forest	61%	
CNN ResNet-18	49%	Near random
Binary (Below 51 / 51+, random baseline 63%)		
XGBoost	**85%+**	Final model of this study
Random Forest	82%	

4.2 Confusion Matrix Analysis

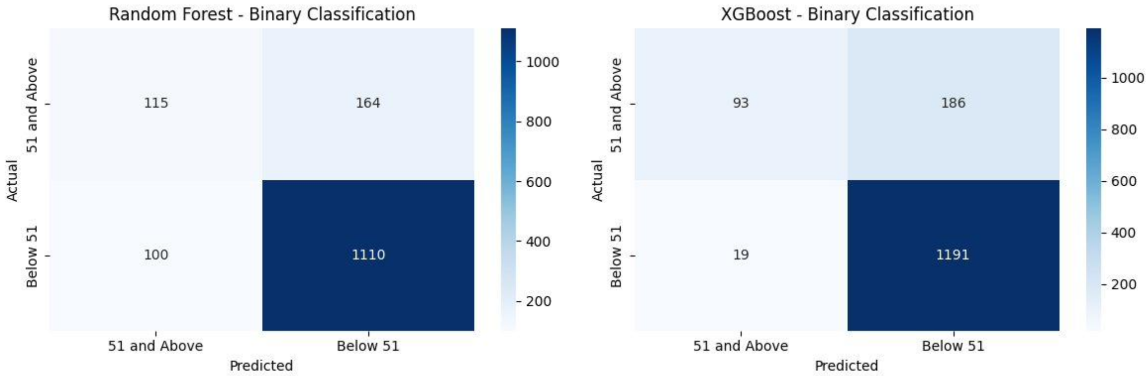

Figure 2: Confusion matrix comparison: Random Forest (left) vs. XGBoost (right) on binary classification. XGBoost Below 51 recall (1191/1210) outperforms RF (1110/1210), while 51+ misclassification rate (186/279) is slightly higher, reflecting the trade-off under class imbalance. Source: plotted by the author, corresponding to `voice_age_binary.ipynb` Cell 8.

Accuracy is high for the Below 51 group, primarily due to the abundant samples (6,048). Partial misclassification of 51+ samples is attributable to class imbalance, emotional speech interference, and individual variation (some older actors retain healthier vocal folds than average).

4.3 CNN Comparison Experiment

ResNet-18 achieved only 49% on the 3-class task due to: insufficient sample size (7,442 samples severely overfitting an 11M-parameter model); 91 actors with repeated utterances yielding far fewer independent samples than 7,442; and Mel spectrograms dominated by emotional interference that overshadows ageing signals. This result **inversely validates** the principle: when domain knowledge is explicit and data are limited, feature engineering + ensemble trees $\succ$ end-to-end deep learning.

4.4 XGBoost Feature Importance (Gain)

Gain measures the average loss reduction brought by a feature at split nodes, defined as follows (adapted from [2]):

$$\text{Gain}(j) = \frac{1}{|T_j|} \sum_{t \in T_j} \left[\mathcal{L}_{\text{before}}(t) - \mathcal{L}_{\text{after}}(t)\right] \tag{3}$$

Top-five Gain features: MFCC_8, MFCC_{11}, HNR, Jitter, MFCC_{12}—highly consistent with SHAP rankings. Gain cannot reflect the **direction** of feature influence; SHAP is therefore more appropriate for medical interpretation.

5 Explainability Analysis (SHAP)

5.1 SHAP Methodology

SHAP[3] was selected as the explainability tool for this study primarily because it decomposes each prediction into individual feature contributions with directionality—that is, it reveals whether a feature's value is *pushing* the prediction towards 51 and Above or towards Below 51. This is especially important here, because what we aim to verify is not merely whether the model is accurate, but whether the model's reasoning corresponds to the physiological mechanisms described in Section 1.

SHAP is grounded in the Shapley value from cooperative game theory, defined as follows (adapted from the original paper[3]):

$$\phi_j(v) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{j\}) - v(S)\big] \tag{4}$$

Intuitively, ϕ_j can be understood as "the average change in the prediction when feature j is added, across all possible feature subsets."

In practice, XGBoost's binary classification output corresponds to two classes in `le.classes_`, one of which is Below 51. To make positive values in the chart intuitively correspond to "pushing towards ageing (51+)", the SHAP values for the Below 51 class are negated before plotting:

```
shap_for_plot = -shap_values[:, :, list(le.classes_).index('Below␣
    51')]
```

After this transformation, a positive SHAP value indicates that the feature pushes the model output towards "51 and Above", while a negative value indicates a push towards "Below 51", facilitating step-by-step verification against the medical mechanisms described in Section 1.

5.2 SHAP Visualisation Results

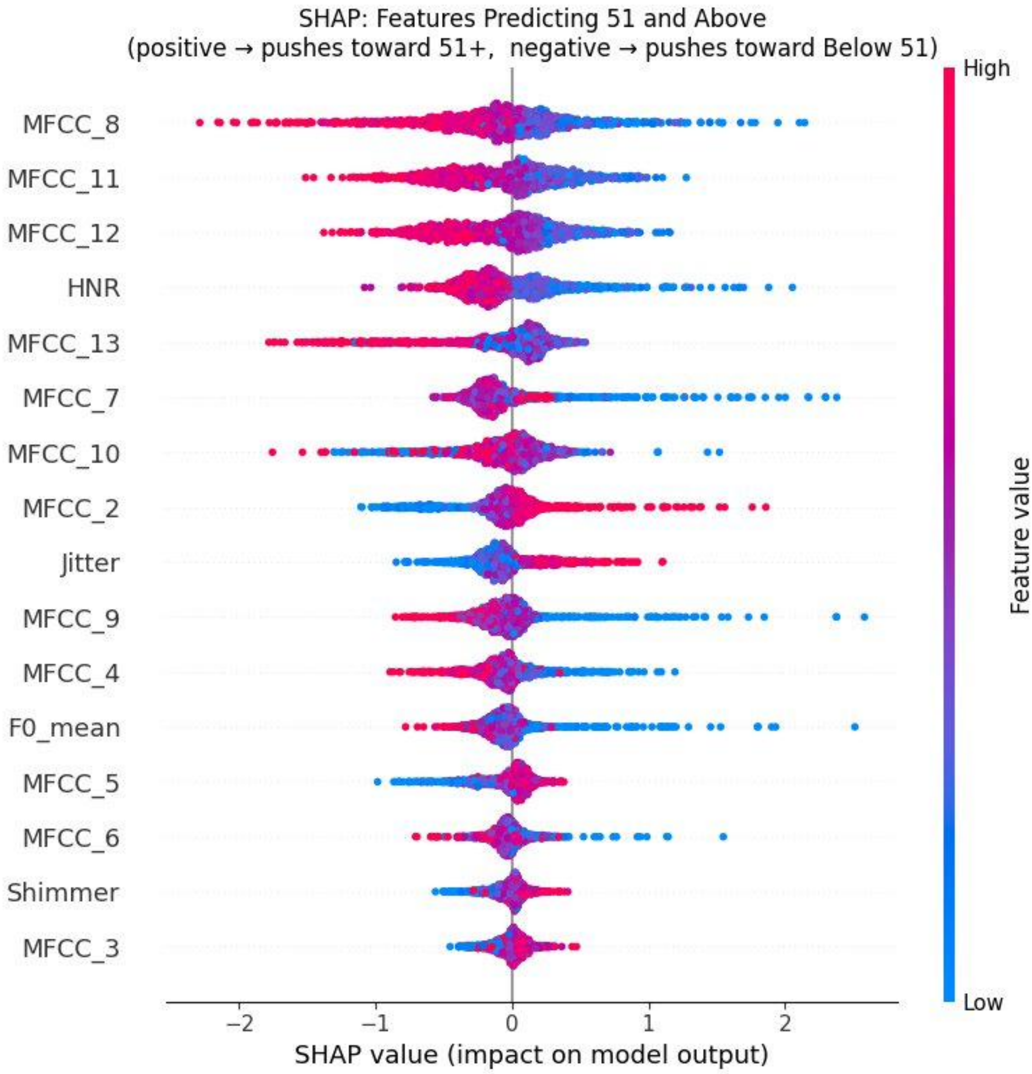

Figure 3: SHAP Beeswarm Plot. The vertical axis ranks features by mean $|\phi_j|$; positive values on the horizontal axis push towards 51+, negative values push towards Below 51; red/blue indicates high/low feature values. Source: plotted by the author, corresponding to `voice_age_binary.ipynb` Cell 9 (SHAP analysis, direction corrected as $-$shap_values[:, :, below_idx]).

5.3 Medical Theory Verification

Table 6: SHAP Results vs. Medical Theory: Comparative Verification

Feature	**SHAP Observation**	**Medical Expectation**	**Verdict**
HNR	Low HNR (blue) pushes towards 51+	Incomplete closure after ageing lowers HNR	✓ Consistent
Jitter	High Jitter (red) pushes towards 51+	Rises after neural deterioration	✓ Consistent
F0_mean	Mixed SHAP direction	Opposite directions by sex	✓ Consistent
Shimmer	SHAP near zero	Professional training masks deterioration	△ Weak signal
$\text{MFCC}_{8\text{–}12}$	Highest-impact feature group	Mid-range MFCCs reflect vocal tract ageing	✓ Plausible

The HNR direction fully aligns with the causal chain "ageing $\rightarrow$ incomplete closure $\rightarrow$ air leakage $\rightarrow$ lower HNR"[4]. The weak Shimmer signal reflects professional vocal training masking age-related deterioration—a dataset selection bias.

6 Interactive Demo System

6.1 System Architecture

This study built an end-to-end interactive demo system using **Gradio**, available at `https://github.com/chelsietao/vocal-age-analyzer`. The data flow is as follows:

Microphone recording / WAV upload ⟶ Feature extraction (16 dimensions) ⟶ XGBoost inference ⟶ SHAP computation ⟶ Pie probability chart + SHAP bar chart + text interpretation

6.2 System Features

- **Input**: Live microphone recording or WAV upload (approx. 3 seconds)
- **Output 1**: Predicted class label and per-class probabilities
- **Output 2**: Pie probability chart (displays prediction confidence)
- **Output 3**: SHAP bar chart (orange: pushes towards 51+; blue-grey: pushes towards Below 51)
- **Output 4**: Clinical interpretation of F0, Jitter, Shimmer, and HNR values
- **Bottom panel**: Background explainer on Presbyphonia, MFCC, and SHAP
- **Bilingual**: Supports Chinese / English language toggle

6.3 User Privacy Statement

The demo system asks users to record or upload approximately 3 seconds of speech in real time. This speech constitutes newly provided biometric data from the user, distinct from the training data (CREMA-D) in terms of privacy implications. The following points are therefore noted explicitly:

- Audio provided by the user is used solely for real-time feature extraction and model inference; once computation is complete it is released from memory and **will not be stored, logged, or used for retraining**.
- The system interface clearly communicates the above data handling policy in both Chinese and English; users may decide for themselves whether to provide a speech sample.
- System output constitutes only a statistical risk indicator and does not constitute a medical diagnosis; this is stated within the interface (see Section 8.1).

7 Limitations and Future Directions

7.1 Current Limitations

Reviewing the full research pipeline from data source through sample structure to model performance reveals that the system's limitations are interconnected—each earlier limitation tends to be a precondition for the next, forming a cascading causal chain.

First, CREMA-D was collected for emotion recognition research; its recording design was never intended to highlight ageing-related acoustic changes, but rather to have actors portray various emotions. This fundamental design difference means that emotional signals become superimposed on ageing signals within the same audio clip. Even though this study selected features theoretically sensitive to ageing, this layer of interference persists and propagates into the downstream sample distribution.

Next, after splitting samples at the age-51 threshold, 51+ samples constitute only 18.7% of the total. This imbalance, rooted in the data collection design, directly limits the model's learning opportunities for the minority class. Even applying `scale_pos_weight` as a compensatory weight adjustment can only modify the loss function calculation within the existing sample distribution; it cannot fundamentally replenish the diversity that the minority class lacks in feature space—and this compounds with the characteristics of the actor sample pool.

Furthermore, all CREMA-D actors are professional performers whose long-term vocal training means their vocal fold health generally surpasses that of age-matched laypersons. As a result, the model's learned representation of ageing within the already sparse 51+ class is further eroded, shifting towards the healthier, less-deteriorated end. When the model is eventually applied to general populations, this dual bias creates a risk of reduced accuracy.

At the same time, because F0 ageing trajectories run in opposite directions by sex, training on a mixed-sex sample is equivalent to superimposing two mutually cancelling signals on the same feature dimension. This unseparated sex effect intertwines with the class imbalance and sample bias described above, jointly compressing the effective information the model can extract from the data.

Finally, returning to the most fundamental constraint: a dataset of 91 individuals remains limited in statistical terms. This scale limitation at the source is precisely why data-hungry methods such as CNN perform poorly on this task, and is the common foundation that amplifies all four preceding limitations.

7.2 Future Research Directions

Building on the interconnected nature of the limitations above, a reasonable path forward would begin with the most fundamental data source issues and then progressively address sample structure and model design constraints.

The most fundamental breakthrough lies in building or acquiring a vocal corpus designed with ageing as its explicit objective—for example, using datasets such as TORGO or SVD, or collecting clinical real-world speech for transfer learning. Only when the training data source is no longer contaminated by emotional speech can the signals carried by the features have a realistic chance of more accurately reflecting genuine vocal ageing.

Once the data source issue is progressively addressed, multi-task learning could be introduced to allow the model to simultaneously predict age group and emotion category within the same training process. This concurrent learning design offers the model an opportunity to learn to disentangle the two signals, thereby reducing emotional speech interference in ageing assessment—and this disentanglement capability would in turn lay the groundwork for personalised analysis.

Once the model has acquired the capacity to separate different signals, obtaining longitudinal recordings from the same individuals at different ages would enable the construction of personalised ageing trajectories. This longitudinal design would transform the original cross-sectional group comparison into a long-term observation of individual physiological change, providing a clinical reference more tailored to individual cases.

As the constraints on speech data are progressively relaxed, multimodal fusion with other physiological signals such as facial images could also be explored. The complementary information from different modalities could provide a more comprehensive basis for ageing assessment—though any such multimodal architecture must ultimately be validated in clinical settings.

Therefore, the final step of the entire research pathway is to collaborate with otolaryngologists, using laryngoscopic examination results as the gold standard to holistically validate the improvements accumulated across all stages— confirming whether the progressively layered design of data sourcing, signal separation, personalised tracking, and multimodal fusion ultimately yields genuine improvements in the model's ability to assess vocal fold ageing.

8 Ethical Considerations and Responsible AI

8.1 Screening Aid, Not a Diagnostic Replacement

This system is positioned as a **screening aid**: its output should be understood as a statistical risk indicator whose purpose is, in a large-scale, low-cost context, to flag individuals whose vocal fold condition may warrant further attention. An ENT specialist would then confirm and diagnose these individuals through laryngoscopy and other professional examinations—forming a progressively converging workflow from preliminary screening to specialist diagnosis. The system interface also explicitly informs users that the training data originates from an emotional speech dataset with known biases, and that the output is for reference only and should not be treated as a medical conclusion.

8.2 Data Privacy and Algorithmic Bias

This study's approach to data privacy can be divided into two levels based on data source—training data and demo system user data—which are independent of each other yet together constitute the full privacy picture for this system.

At the training data level, CREMA-D is a public academic dataset whose licence permits non-commercial academic research. This study uses only the acoustic features derived via feature extraction, does not involve any additional identification of actors as individuals, and will not redistribute original audio files. This handling ensures that use of the training data remains within the permitted scope at all times.

At the demo system user level, speech collected live constitutes biometric data potentially protected under regulations such as GDPR and Taiwan's Personal Data Protection Act. To address this risk, the demo system performs local real-time computation; user-provided speech is released immediately after inference and is not stored. This approach was described in Section 6.3 and is displayed within the interface, enabling users to understand how their data will be handled before recording.

Beyond these two levels, the training data itself carries several potential biases that propagate downward from the data source and affect the model's generalisability: demographic bias from samples predominantly comprising North American native English speakers; occupational bias from professional actors whose vocal fold health generally exceeds that of the general population; and gender bias from potentially unbalanced male-to-female ratios and age group distributions. These biases correspond to the limitations described in Section 7.1 and collectively constitute risk sources that must be continuously monitored when the model is deployed in practice.

8.3 Explainability as the Foundation of Trust

The role of SHAP in this study can be understood as a bridge connecting model outputs to medical theory, allowing every prediction to be decomposed into individual feature contributions and mapped one-by-one to the medically verifiable physiological mechanisms described in Section 1, rather than remaining a single output value. When SHAP shows that a low HNR pushes the prediction towards 51 and Above, a clinician can draw on their specialist knowledge of Presbyphonia to independently assess whether this correspondence is reasonable—and to raise objections or counterarguments when necessary. This capacity for layer-by-layer verification by domain experts is the core of building **evidence-based trust**, and forms an important foundation for the trustworthy deployment of AI systems in medical settings.

9 Conclusion

This study successfully built a binary classification system for vocal cord physiological age. The main contributions are as follows:

1. **High accuracy**: XGBoost achieved 85%+, a 22-percentage-point improvement over the random baseline.
2. **Explainability verification**: SHAP confirmed that AI decisions align with Presbyphonia medical literature, particularly validating the central roles of decreased HNR and increased Jitter.
3. **Methodological comparison**: A three-model comparison verified that, on small-scale data, feature engineering + ensemble trees $\succ$ deep learning.
4. **Application demonstration**: The bilingual Gradio system provides a proof of concept for non-invasive vocal fold screening.

This study embodies the core spirit of **AI for Science**: not merely pursuing accuracy, but emphasising explainability and fidelity to medical theory, laying a foundation for the trustworthy deployment of medical AI.

References

References

[1] Cao, H., Cooper, D.G., et al. (2014). *CREMA-D: Crowd-sourced Emotional Multimodal Actors Dataset*. IEEE Trans. Affective Computing, 5(4), 377–390.

[2] Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD 2016), 785–794.

[3] Lundberg, S.M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. Advances in Neural Information Processing Systems 30 (NeurIPS 2017), 4765–4774.

[4] Kendall, K. (2007). *Presbyphonia: A Review*. Current Opinion in Otolaryngology & Head and Neck Surgery, 15(3), 137–140.

[5] Teixeira, J.P., Oliveira, C., & Lopes, C. (2013). *Vocal Acoustic Analysis—Jitter, Shimmer and HNR Parameters*. Procedia Technology, 9, 1112–1122.

Further Reading / Online Resources

The following links are supplementary popular-science and technical references consulted during writing. They differ in nature from the academic citations above [1, 2, 3, 4, 5] and are therefore listed separately without formal citation numbers.

Machine Learning Algorithms:

- IBM: What is XGBoost? `https://www.ibm.com/topics/xgboost`
- IBM: What is Random Forest? `https://www.ibm.com/topics/random-forest`

Explainable AI:

- IBM: What is Explainable AI? `https://www.ibm.com/topics/explainable-ai`
- SHAP GitHub project page (documentation and examples): `https://github.com/shap/shap`

Acoustic Features and Speech Processing:

- Wikipedia: Mel-frequency cepstrum (MFCC): `https://en.wikipedia.org/wiki/Mel-frequency_cepstrum`
- Wikipedia: Jitter (signal processing): `https://en.wikipedia.org/wiki/Jitter`

Vocal Medicine Background:

- Mayo Clinic: Voice disorders overview: `https://www.mayoclinic.org/diseases-conditions/voice-disorders/symptoms-causes/syc-20353022`
- Merck Manual: Laryngeal Disorders: `https://www.merckmanuals.com/professional/ear,-nose,-and-throat-disorders/laryngeal-disorders`

Appendix A: Generative AI Collaboration Disclosure

Generative AI (Claude, Anthropic) was used as a collaborative tool during the production of this work. The scope and estimated proportions are described below:

- **Completed independently by the author (core research content)**: Research topic and problem definition (including the choice of age 51 as threshold); feature engineering direction (including the rationale for removing MFCC_1); experimental design and version evolution planning; analysis and interpretation of model training results; comparative verification of SHAP results against medical theory; planning of research limitations and future directions; and final editing and quality control of the full manuscript.
- **AI-assisted portions (estimated by output type)**:
  - **Code**: Of approximately 600–800 lines across the full project (comprising four notebooks: 3-class training, binary classification, CNN comparison, and Gradio demo), AI assistance with debugging, syntax correction, and performance optimisation accounts for approximately 20–25%. This was concentrated mainly in correcting the SHAP direction formula, XGBoost hyperparameter tuning suggestions, and Gradio interface layout and bilingual toggle logic. The core feature engineering logic, model selection, and evaluation pipeline design were written by the author.
  - **Written report**: The full report spans approximately 16 pages across nine sections. AI collaboration was concentrated in text polishing, paragraph structure adjustment, and LaTeX typesetting (approximately 10–15% of total content). The research content, data analysis, SHAP and medical theory comparison, and the argumentative threads of Sections 1, 7, and 8 were all drafted by the author before AI was used for proofreading and typesetting adjustments.
  - **Supplementary documents**: For the README (bilingual) and presentation outline, AI collaboration was proportionally higher (approximately 50% or above), primarily because these documents are informational and format-conversion in nature, drawing content from the completed research in the main report.
- **Disclaimer**: The proportions above are the author's retrospective estimates and are not precise statistics. All AI-assisted content was reviewed, understood, and revised by the author before adoption. The core judgements, data analysis results, and conclusions are the author's own academic work; the author bears full academic and authorial responsibility for the entire manuscript.